```
n1 = 1;
n2 = 1;
g1 = 3;
g2 = 1;
n2bar = 1;
g2bar = 1;

(*Consistencies solution for the choice of pentaquark *)
γ12 = 0.3954232044351274;
(*γ12 = 0.3954613003182025;*)(*relation between light and charm quark*)
(*β12bar =  0.1408736554002409; *)
β12bar = 0.2753036090258285;
(* relation between light and anti-charm quark *)
(*β22bar= 0.9849242236403603; *)
(* relation between charm and anti-charm quark *)
β22bar = 0.9621573728871041;


(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
bb = 1553;
aa = zz / bb; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n2 * g2 + n2bar * g2bar) / ;
```

$$\text{theta} = \frac{-6\,\eta}{5\,(5\,\eta - 1)};$$ (*Just a multiplication factor for a particular value of eta *)

Q1 = −0.45;

(*Q1=theta*3*$\frac{(n1*g1-1)}{4n1}$;*)(*Q1 is the constant in the differential equation $d^2\frac{f1}{dx^2}$=

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

Q2 = −0.851982694608053;

(*Q2= 3*theta*((n1*g1−1)*γ12^1.5/n1+g2)/(4*γ12);*)

(*Q2 is the constant in the differential equation $d^2\frac{f2}{dx^2}$=

Q2 $\frac{f2^{\frac{3}{2}}}{\sqrt{x}}$ in the middle region. This is obtained by solving

two consistency conditions along with other constant γ12. *)

(*Q2bar = theta*(g1*β12bar^1.5+g2*β22bar^1.5)/alfa2 ;*)

Q2bar = −2.0971474564888006;

(*Qbar is the constant in the differential equation $d^2\frac{f3}{dx^2}$=

Q2bar $\frac{f3^{\frac{3}{2}}}{\sqrt{x}}$ in the inner region. This is obtained by solving three

consistency conditions along with other constants β13 & β23. *)

g = alfas;(*Why g = alfas in this case ?*)

(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication

facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)

$$R = \frac{3\,a}{(8 * \text{alfas})} * \left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{2}{3}};$$(*Page number 54, it is inverse in the copy though *)

$$c = \frac{12}{5\,\pi}\,\frac{4}{3}\,g^2\,\frac{4}{3}\,\frac{\text{alfas}}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}};$$

(*Multiplication factor in the expression of kinetic energy *)

$$d = \frac{4}{3}\,g^2\,\frac{4}{3}\,\frac{\text{alfas}}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}}\,\frac{18}{25\,\pi}\,\frac{\eta}{\left(\eta - \frac{1}{5}\right)};$$ (* $\frac{18g^2}{5(5\eta\ -1\ )}$.16 π^2.$\left(\frac{3a}{8\text{alfas}}\right)$. $\frac{R^2}{(6\pi^2)^2}$,

The value of R is from page number 54 which is written inverse here,

and $\left(\frac{3a}{8\text{alfas}}\right)$ is inverse in copy, BA, 13 September *)

(*Multiplication factor in the expression of Potential energy *)

```
(*Let us define the three functions for three regions *)

f2[x2_?NumericQ, x2bar_?NumericQ, u2bar_?NumericQ , s2bar_?NumericQ] :=
  Abs[x2 * y2'[x2] - y2[x2] - n2 * Q2/(3 η) - β22bar * n2bar * Q2bar/(3 η) + β22bar^(3/2) * n2bar * Q2/(3 η)]
   (*Appendix from Kinetic energy*) /.
   First@NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)])/x^(1/2), y2'[x2bar] == β22bar * s2bar ,
      y2[x2bar] == β22bar * u2bar (* s2bar is the first derivative of f_2bar(x) *)
      (* u2bar is the function f_2bar(x) *)(* It shows how the
       function f2(x2) changes on other regions*)}, y2, {x, x2bar, x2}];
(*This is the function f2 defined on the middle region. It has
  the differential equation  d^2 f2/dx^2 =
 Q2 f2^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x2_?NumericQ, s2_?NumericQ, u2_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12 * n2 * Q2/(3 η) +
     γ12^(3/2) * n2 * Q1/(3 η) - (γ12^(3/2) * β22bar^(3/2) - β12bar^(3/2)) * n2bar * Q1/(3 η) +
     γ12 * β22bar^(3/2) * n2bar * Q2/(3 η) - γ12 * β22bar * n2bar * Q2bar/(3 η)]
    (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)])/x^(1/2),
      y1'[x2] == γ12 * s2, (* It means at the boundary x2
       the function df1 changes to other function *)
      y1[x2] == γ12 * u2 (* s2 is the first derivative of f_2(x) *)
      (* u2 is the function f_2(x) *)
      (*It means at the boundary x2 the function f1 changes to other function*)
     }, (* It means at the boundary x3 the function f1 changes to other function *)
```

```
    y1, {x, x2, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
 Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
    condition of d/dx f1(x)|x2 = γ12 d/dx f2(x)|x2 and f1(x2) = γ12 f2(x2)  *)

f2bar[xmax_?NumericQ, m2_?NumericQ] :=
  Module[{xx = xmax},
   solution2bar = NDSolve[{y2bar''[x] == (Q2bar * Abs[y2bar[x]]^(3/2))/x^(1/2),
      y2bar'[1.*^-9] == m2, y2bar[1.*^-9] == 0.}, y2bar, {x, 1.*^-9, xx}];

   m = NIntegrate[Sqrt[x] * Evaluate[y2bar[x] /. solution2bar]^(3/2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2bar / (3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x3 inner region. On this region,
f1(x)=β13 f3(x) and f2(x)= β23 f3(x).   *)



(*Start the program*)

m2 = 9;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)
```

```
mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
 m2 = mqn;
 topi = 0.01;
 m2B = m2;
 Energy = 10^9;
 Energy = 10^9;
 Energy2plus = 10^9;
 H = 10^9 + 1;
 SN = 0;
 nloop = 1;
 For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
   H = Energy;
   HX = Energy6bare;
   HY = Energy2plus; m2 = m2 + SN * 10^(-n);
   x2bar = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
   FindMinimum[f2bar[x2bar, m2], {x2bar, topi}];
   (*remove this function and investigate if it is actually required BS 4/3/2021*)
   x2bar = x2bar /. Last[FindMinimum[f2bar[x2bar, m2], {x2bar, topi}]];
   solution2bar = NDSolve[{y2bar''[x] == (Q2bar * Abs[y2bar[x]^(3/2)]) / x^(1/2),
       y2bar'[1.*^-9] == m2, y2bar[1.*^-9] == 0.}, y2bar, {x, 1.*^-9, x2bar}];
   t1 = Evaluate[y2bar[x2bar] /. solution2bar];
   t4 = Evaluate[y2bar'[x2bar] /. solution2bar];
   u2bar = t1[[1]];
   (* u2bar is the function f_2bar(x) *)
   s2bar = t4[[1]];
   (* s2bar is the first derivative of f_2bar(x) *)
   If[u2bar < 0., u2bar = 0.];


   x2 = x2bar + 0.1;
   FindMinimum[{f2[x2, x2bar,  u2bar, s2bar]}, {x2, x2bar}];
   x2 = x2 /. Last[FindMinimum[{f2[x2, x2bar,  u2bar, s2bar]}, {x2, x2bar}]];
   solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2),
```

```
    y2'[x2bar] == β22bar * s2bar , y2[x2bar] == β22bar * u2bar}, y2, {x, x2bar, x2}];
t2 = Evaluate[y2[x2] /. solution2];
t5 = Evaluate[y2'[x2] /. solution2];
u2 = t2[[1]];
(* u2 is the function f_2(x) *)
s2 = t5[[1]];
(* s2 is the first derivative of f_2(x) *)
If[u2 < 0., u2 = 0.];
x1 = x2 + 0.1;
FindMinimum[{f1[x1, x2, s2, u2]}, {x1, x2}];
x1 = x1 /. Last[FindMinimum[{f1[x1, x2, s2, u2]}, {x1, x2}]];
solution1 = NDSolve[
  {y1''[x] == (Q1 * Abs[y1[x]^(3/2)]) / x^(1/2), y1'[x2] == γ12 * s2 , y1[x2] == γ12 * u2 }, y1, {x, x2, x1}];
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]];
(* u1 is the function f_1(x) *)
s1 = t6[[1]];
(* s1 is the first derivative of f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)
h8 = NIntegrate[Evaluate[y2bar[x] /. solution2bar]^(3/2) * √x,
  {x, 0, x2bar}, Method → {Automatic, "SymbolicProcessing" → 0}];
h9 = NIntegrate[Evaluate[y2bar[x] * β22bar /. solution2bar]^(3/2) * √x,
  {x, 0, x2bar}, Method → {Automatic, "SymbolicProcessing" → 0}];
h10 = NIntegrate[Evaluate[y2[x] /. solution2]^(3/2) * √x, {x, x2bar, x2},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h11 = NIntegrate[Evaluate[β12bar * y2bar[x] /. solution2bar]^(3/2) * √x,
  {x, 0, x2bar}, Method → {Automatic, "SymbolicProcessing" → 0}];
h12 = NIntegrate[Evaluate[γ12 * y2[x] /. solution2]^(3/2) * √x, {x, x2bar, x2},
```

```
  Method → {Automatic, "SymbolicProcessing" → 0}];
h13 = NIntegrate[Evaluate[y1[x] /. solution1]^(3/2) * √x, {x, x2, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];

h20 = h8[[1]];
(*normalization condition for heavy anti-charm quark*)

h14 = h9 + h10 ;
h30 = h14[[1]]; (*normalization condition for heavy charm quark *)

h15 = h11 + h12 + h13;
h40 = h15[[1]]; (*normalization condition for light quark*)
```

$$\text{normx2barr} = \text{x2bar}\ \text{s2bar} - \text{u2bar}\ ;$$

$$\text{normx2bar} = \frac{\text{n2bar}\ \text{Q2bar}}{3\,\eta}\ ;$$

```
(*normx2barr and normx2bar should be equal *)
normmx2 = x2 s2 - u2 ;
```

$$\text{normx2} = \frac{\text{n2}\ \text{Q2}}{3\,\eta} + \beta 22\text{bar}\,\frac{\text{n2bar}\ \text{Q2bar}}{3\,\eta} - (\beta 22\text{bar})^{\frac{3}{2}}\,\frac{\text{n2bar}\ \text{Q2}}{3\,\eta}\ ;$$

```
(*normmx2 and normx2 should be equal *)
normmx1 = x1 s1 - u1 ;
```

$$\text{normx1} = \frac{\text{n1}\ \text{Q1}}{3\,\eta} + \gamma 12\,\frac{\text{n2}\ \text{Q2}}{3\,\eta} - (\gamma 12)^{\frac{3}{2}}\,\frac{\text{n2}\ \text{Q1}}{3\,\eta} + \left((\gamma 12)^{\frac{3}{2}}\,(\beta 22\text{bar})^{\frac{3}{2}} - (\beta 12\text{bar})^{\frac{3}{2}}\right)\frac{\text{n2bar}\ \text{Q1}}{3\,\eta} - \gamma 12\,(\beta 22\text{bar})^{\frac{3}{2}}\,\frac{\text{n2bar}\ \text{Q2}}{3\,\eta} + \gamma 12\,\beta 22\text{bar}\,\frac{\text{n2bar}\ \text{Q2bar}}{3\,\eta}\ ;$$

```
(*normmx1 and normx1 should be equal *)

h1 = NIntegrate[Evaluate[y2bar[x] /. solution2bar]^2.5 / (x^0.5),
  {x, 0, x2bar}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5), {x, x2bar, x2},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x2, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
```

```
h4 = h1[[1]];
h5 = h2[[1]];
h6 = h3[[1]] ;
AB = 4/7 Sqrt[x2bar] (u2bar)^(5/2) - 5 n2bar/(21 η) s2bar ;
CD = 4/7 Sqrt[x2] (u2)^(5/2) - 4/7 (β22bar)^(5/2) Sqrt[x2bar] (u2bar)^(5/2) -
   5/(7 Q2) s2 (n2 Q2/(3 η) + β22bar n2bar Q2bar/(3 η) - (β22bar)^(3/2) n2bar Q2/(3 η)) +
   5/(7 Q2) (β22bar)^2 n2bar Q2bar/(3 η) s2bar ;
EF = 4/7 Sqrt[x1] (u1)^(5/2) - 4/7 (γ12)^(5/2) Sqrt[x2] (u2)^(5/2) -
   5/(7 Q1) s1 (n1 Q1/(3 η) + γ12 n2 Q2/(3 η) - (γ12)^(3/2) n2 Q1/(3 η) + n2bar Q1/(3 η) ((γ12)^(3/2) (β22bar)^(3/2) - (β12bar)^(3/2)) -
      (γ12) (β22bar)^(3/2) n2bar Q2/(3 η) + (γ12) (β22bar) n2bar Q2bar/(3 η)) +
   5/(7 Q1) (γ12)^2 s2 (n2 Q2/(3 η) + β22bar n2bar Q2bar/(3 η) - (β22bar)^(3/2) n2bar Q2/(3 η));
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)

K2bar2barr = n2bar/(3 η Q2bar) s2bar - 1/Q2bar h4 ;
K2bar2bar = 4 n2bar/(7 η Q2bar) s2bar - 4/(7 Q2bar) Sqrt[x2bar] (u2bar)^(5/2);
(*K2bar2barr and K2bar2bar should be equal *)
K2bar2 = -4/(7 Q2bar) β22bar^1.5 Sqrt[x2bar] (u2bar)^(5/2) +
```

$$\frac{n2bar}{3\,\eta}\left(\frac{\beta22bar^{1.5}}{Q2bar}-\frac{\beta22bar}{Q2}+\frac{5*\beta22bar^{1.5}}{7\,Q2bar}\right)s2bar+\frac{n2bar}{3\,\eta\,Q2}s2\;;$$

$$K2barr2=\frac{n2bar\,(\beta22bar)^{\frac{3}{2}}}{3\,\eta\,Q2bar}s2bar-\frac{n2bar\,\beta22bar}{3\,\eta\,Q2}s2bar+\frac{n2bar}{3\,\eta\,Q2}s2-\frac{(\beta22bar)^{\frac{3}{2}}}{Q2bar}h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K2bar1=-\frac{4}{7\,Q2bar}\beta12bar^{1.5}\sqrt{x2bar}\;(u2bar)^{\frac{5}{2}}+$$

$$\frac{n2bar}{3\,\eta}\left(\frac{5}{7\,Q2bar}\beta12bar^{1.5}+\frac{\beta12bar^{1.5}}{Q2bar}-\frac{\gamma12^{1.5}*\beta22bar}{Q2}\right)s2bar+$$

$$\frac{n2bar}{3\,\eta}\left(\frac{\gamma12^{1.5}}{Q2}-\frac{\gamma12}{Q1}\right)s2+\frac{n2bar}{3\,\eta\,Q1}s1\;;$$

$$K2barr1=\frac{n2bar}{3\,\eta\,Q2bar}(\beta12bar)^{\frac{3}{2}}s2bar+\frac{n2bar\,(\gamma12)^{\frac{3}{2}}}{3\,\eta\,Q2}s2-$$

$$\frac{n2bar}{3\,\eta\,Q2}(\gamma12)^{\frac{3}{2}}\beta22bar\,s2bar+\frac{n2bar}{3\,\eta\,Q1}s1-\frac{n2bar}{3\,\eta\,Q1}\gamma12\,s2-\frac{(\beta12bar)^{\frac{3}{2}}}{Q2bar}h4\;;$$

(*K2bar1 and K2barr1 should be equal *)

$$K12=\frac{4}{7}\beta22bar^{1.5}\left(\frac{\gamma12^{1.5}*\beta22bar}{Q2}-\frac{\beta12bar^{1.5}}{Q2bar}\right)\sqrt{x2bar}\;(u2bar)^{\frac{5}{2}}-$$

$$\frac{4}{7\,Q2}\gamma12^{1.5}\sqrt{x2}\;(u2)^{\frac{5}{2}}+\frac{n2bar}{3\,\eta}\beta22bar$$

$$\left(\frac{5}{7\,Q2bar}\beta22bar^{\frac{1}{2}}*\beta12bar^{1.5}-\frac{\gamma12^{1.5}*\beta22bar^{1.5}}{Q2}-\frac{5\,Q2bar}{7\,Q2^2}\gamma12^{1.5}*\beta22bar+\right.$$

$$\left.\frac{Q2bar}{Q2^2}\gamma12^{1.5}*\beta22bar+\frac{1}{Q2bar}\beta12bar^{1.5}*\beta22bar^{\frac{1}{2}}-\frac{\beta12bar^{1.5}}{Q2}\right)s2bar+$$

$$\frac{1}{3\,\eta\,Q2}\left(n2 * \gamma 12^{1.5} - \frac{n2\,Q2}{Q1}\,\gamma 12 + n2bar * \beta 12bar^{1.5} - \frac{n2bar\,Q2bar}{Q2}\,\gamma 12^{1.5} * \beta 22bar + \frac{5}{7}\,\gamma 12^{1.5}\left(n2 + \frac{n2bar\,Q2bar}{Q2}\,\beta 22bar - n2bar * \beta 22bar^{1.5}\right)\right)s2 + \frac{n2}{3\,\eta\,Q1}\,s1 ;$$

```
K122 = n2bar / (3 η Q2) (β12bar) ^ (3 / 2) s2 - n2bar / (3 η Q2) (β12bar) ^ (3 / 2) β22bar s2bar -
   ((β12bar) ^ (3 / 2) (β22bar) ^ (3 / 2)) / Q2bar h4 - (γ12) ^ (3 / 2) / Q2 h5 - (n2bar Q2bar) / (3 η Q2 ^ 2)
    (γ12) ^ (3 / 2) β22bar s2 + (n2bar Q2bar) / (3 η Q2 ^ 2) (γ12) ^ (3 / 2) (β22bar) ^ 2 s2bar +
   n2bar / (3 η) (β22bar) ^ (3 / 2) (β12bar ^ (3 / 2) / Q2bar - ((γ12) ^ (3 / 2) β22bar) / Q2) s2bar +
   n2 / (3 η) γ12 (γ12 ^ (1 / 2) / Q2 - 1 / Q1) s2 + n2 / (3 η Q1) s1 ;
(*K12 and K122 should be equal*)
```

$$K22 = \frac{4}{7}\,\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q2} - \frac{\beta 22bar}{Q2bar}\right)\sqrt{x2bar}\;(u2bar)^{\frac{5}{2}} - \frac{4}{7\,Q2}\sqrt{x2}\;(u2)^{\frac{5}{2}} + \frac{n2bar}{3\,\eta}\,\beta 22bar^2\left(\beta 22bar\,\frac{12}{7\,Q2bar} + \frac{2}{7}\,\frac{Q2bar}{Q2^2} - \frac{2 * \beta 22bar^{\frac{1}{2}}}{Q2}\right)s2bar + \left(\frac{12\,n2}{21\,\eta\,Q2} + \beta 22bar\,\frac{n2bar}{3\,\eta\,Q2}\left(\beta 22bar^{\frac{1}{2}} - \frac{2}{7}\,\frac{Q2bar}{Q2} - \frac{5}{7}\,\beta 22bar^{\frac{1}{2}}\right)\right)s2 ;$$

$$K222 = \frac{n2bar\,Q2bar}{3\,\eta\,Q2}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2bar} - \frac{\beta 22bar}{Q2}\right)s2 - \frac{n2bar\,Q2bar * \beta 22bar}{3\,\eta\,Q2}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2bar} - \frac{\beta 22bar}{Q2}\right)s2bar - \frac{(\beta 22bar)^3}{Q2bar}\,h4 - \frac{1}{Q2}\,h5 + (\beta 22bar)^3\,\frac{n2bar}{3\,\eta\,Q2bar}\,s2bar - (\beta 22bar)^{\frac{5}{2}}\,\frac{n2bar}{3\,\eta\,Q2}\,s2bar + \frac{n2}{3\,\eta\,Q2}\,s2 ;$$

```
(*K22 and K222 should be equal *)
```

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta 22bar^{\frac{5}{2}} * \gamma 12^3}{Q2} - \frac{\beta 12bar^3}{Q2bar}\right)\sqrt{x2bar}\,(u2bar)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma 12^{\frac{5}{2}}}{Q1} - \frac{\gamma 12^3}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} - \frac{4}{7\,Q1}\sqrt{x1}\,(u1)^{\frac{5}{2}} + \left(\frac{n2bar}{3\,\eta}\left(\frac{\beta 12bar^3}{Q2bar} - \frac{\beta 12bar^{\frac{3}{2}} * \gamma 12^{\frac{3}{2}} * \beta 22bar}{Q2} + \frac{Q2bar}{Q2^2}\gamma 12^3 * \beta 22bar^2 - \frac{\beta 12bar^{1.5} * \gamma 12^{1.5} * \beta 22bar}{Q2}\right) + \frac{5\,n2bar}{21\,\eta}\left(\frac{\beta 12bar^3}{Q2bar} - \frac{Q2bar}{Q2^2}\beta 22bar^2 * \gamma 12^3\right)\right)s2bar + \left(\frac{n2bar}{3\,\eta}\left(\frac{\beta 12bar^{1.5} * \gamma 12^{1.5}}{Q2} - \frac{\beta 12bar^{1.5} * \gamma 12}{Q1} - \frac{\beta 22bar^{1.5} * \gamma 12^3}{Q2} + \frac{\beta 22bar^{1.5} * \gamma 12^{\frac{5}{2}}}{Q1} + \frac{\beta 12bar^{1.5} * \gamma 12^{1.5}}{Q2} - \frac{Q2bar}{Q2^2}\gamma 12^3 * \beta 22bar - \frac{\beta 12bar^{1.5} * \gamma 12}{Q1} + \frac{\gamma 12^{\frac{5}{2}} * \beta 22bar^{1.5}}{Q1} + \frac{Q2bar}{Q1^2}\gamma 12^2 * \beta 22bar - \frac{Q2}{Q1^2}\beta 22bar^{1.5} * \gamma 12^2\right) + \frac{n2}{3\,\eta}\left(\frac{\gamma 12^3}{Q2} - \frac{2 * \gamma 12^{\frac{5}{2}}}{Q1} + \frac{Q2}{Q1^2}\gamma 12^2\right) + \frac{5}{21\,\eta}\left(\frac{n2}{Q2}\gamma 12^3 + \frac{n2bar\ Q2bar}{Q2^2}\gamma 12^3 * \beta 22bar - \frac{n2bar}{Q2}\beta 22bar^{1.5} * \gamma 12^3 - \frac{n2\ Q2}{Q1^2}\gamma 12^2 - \frac{n2bar\ Q2bar}{Q1^2}\gamma 12^2 * \beta 22bar + \frac{n2bar\ Q2}{Q1^2}\gamma 12^2 * \beta 22bar^{1.5}\right)\right) * s2 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\gamma 12 + \frac{n2}{3\,\eta\,Q1}\gamma 12^{1.5} + \frac{n2bar}{3\,\eta\ Q1}\left(\beta 12bar^{1.5} - \gamma 12^{1.5} * \beta 22bar^{1.5} - \frac{Q2bar}{Q1}\beta 22bar * \gamma 12 + \frac{Q2}{Q1}\beta 22bar^{1.5} * \gamma 12\right) +$$

$$\frac{5}{21\,\eta}\left(\frac{n1}{Q1}+\frac{n2\ Q2}{Q1^2}\,\gamma12-\frac{n2}{Q1}\,\gamma12^{1.5}+\frac{n2bar}{Q1}\,\gamma12^{1.5}*\beta22bar^{1.5}-\frac{n2bar}{Q1}\,\beta12bar^{1.5}-\frac{n2bar\ Q2}{Q1^2}\,\beta22bar^{1.5}*\gamma12+\frac{n2bar\ Q2bar}{Q1^2}\,\gamma12\ \beta22bar\right)*s1;$$

$$K111=\frac{-(\beta12bar)^3}{Q2bar}\,h4-\frac{(\gamma12)^3}{Q2}\,h5-\frac{1}{Q1}\,h6+\frac{n2bar}{3\,\eta\,Q2}(\beta12bar)^{\frac{3}{2}}(\gamma12)^{\frac{3}{2}}s2-\frac{n2bar\,Q2bar}{3\,\eta\,Q2^2}(\gamma12)^3\,\beta22bar\,s2-\frac{n2bar}{3\,\eta\,Q2}(\beta12bar)^{\frac{3}{2}}(\gamma12)^{\frac{3}{2}}\beta22bar\,s2bar+\frac{n2bar\,Q2bar}{3\,\eta\,Q2^2}(\gamma12)^3(\beta22bar)^2\,s2bar+\left(\frac{n2bar}{3\,\eta\,Q1}(\beta12bar)^{\frac{3}{2}}+\frac{n2}{3\,\eta\,Q1}(\gamma12)^{\frac{3}{2}}-\frac{n2\,Q2}{3\,\eta\,Q1^2}\gamma12-\frac{n2bar}{3\,\eta\,Q1}(\gamma12)^{\frac{3}{2}}(\beta22bar)^{\frac{3}{2}}-\frac{n2bar\,Q2bar}{3\,\eta\,Q1^2}\beta22bar\,\gamma12+\frac{n2bar\,Q2}{3\,\eta\,Q1^2}(\beta22bar)^{\frac{3}{2}}\gamma12\right)s1-\left(\frac{n2bar}{3\,\eta\,Q1}(\beta12bar)^{\frac{3}{2}}\gamma12+\frac{n2}{3\,\eta\,Q1}(\gamma12)^{\frac{5}{2}}-\frac{n2\,Q2}{3\,\eta\,Q1^2}(\gamma12)^2-\frac{n2bar}{3\,\eta\,Q1}(\gamma12)^{\frac{5}{2}}(\beta22bar)^{\frac{3}{2}}-\frac{n2bar\,Q2bar}{3\,\eta\,Q1^2}(\gamma12)^2\,\beta22bar+\frac{n2bar\,Q2}{3\,\eta\,Q1^2}(\beta22bar)^{\frac{3}{2}}(\gamma12)^2\right)s2+$$

```
n2bar / (3 η) (β12bar)^3 / Q2bar s2bar - n2bar / (3 η Q2) (β12bar)^(3 / 2)
 (γ12)^(3 / 2) β22bar s2bar + n2bar / (3 η Q2) (β12bar)^(3 / 2) (γ12)^(3 / 2) s2 -
n2bar / (3 η Q1) (β12bar)^(3 / 2) γ12 s2 + n2 / (3 η Q2) (γ12)^3 s2 -
n2bar / (3 η Q2) (β22bar)^(3 / 2) (γ12)^3 s2 - n2 / (3 η Q1) (γ12)^(5 / 2) s2 +
n2bar / (3 η Q1) (β22bar)^(3 / 2) (γ12)^(5 / 2) s2 + n1 / (3 η Q1) s1;
(*K11 and K111 should be equal *)
```

$$T1=c\left(\frac{4}{7}\left(alfa1*g1\left(\beta12bar^{\frac{5}{2}}-\gamma12^{\frac{5}{2}}*\beta22bar^{\frac{5}{2}}\right)+alfa2*g2bar\right)\sqrt{x2bar}\,(u2bar)^{\frac{5}{2}}+\frac{4}{7}alfa2*g2\sqrt{x2}\,(u2)^{\frac{5}{2}}+\frac{4}{7}alfa1*g1\sqrt{x1}\,(u1)^{\frac{5}{2}}+\frac{5\,n2bar}{21\,\eta}\left(\beta22bar^2\,\frac{Q2bar}{Q2}\left(alfa1*g1*\gamma12^{\frac{5}{2}}+alfa2*g2\right)-\right.\right.$$

$$\left(\text{alfa1} * \text{g1} * \beta 12\text{bar}^{\frac{5}{2}} - \text{alfa2} * \text{g2} * \beta 22\text{bar}^{\frac{5}{2}} - \text{alfa2} * \text{g2bar}\right) \text{s2bar} +$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12^2 * \text{alfa1} * \text{g1}}{\text{Q1}}\left(\text{n2} * \text{Q2} + \beta 22\text{bar} * \text{n2bar} * \text{Q2bar} - \beta 22\text{bar}^{1.5} * \text{n2bar} * \text{Q2}\right) -\right.$$

$$\left(\text{alfa1} * \text{g1} * \gamma 12^{\frac{5}{2}} + \text{alfa2} * \text{g2}\right)$$

$$\left.\left(\text{n2} + \beta 22\text{bar}\,\frac{\text{n2bar} * \text{Q2bar}}{\text{Q2}} - \beta 22\text{bar}^{1.5} * \text{n2bar}\right)\right) \text{s2} -$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1} + \gamma 12\,\frac{\text{n2}\,\text{Q2}}{\text{Q1}} + \gamma 12^{1.5}\left(\beta 22\text{bar}^{1.5} * \text{n2bar} - \text{n2}\right) -\right.$$

$$\left.\beta 12\text{bar}^{1.5} * \text{n2bar} - \gamma 12 * \beta 22\text{bar}^{1.5}\,\frac{\text{n2bar}\,\text{Q2}}{\text{Q1}} + \gamma 12\,\beta 22\text{bar}\,\frac{\text{n2bar}\,\text{Q2bar}}{\text{Q1}}\right) \text{s1}\Bigg);$$

$$\text{V1} = -\text{d}\left(\frac{(\text{g1}\,\text{n1} - 1)}{\text{n1}}\,\text{g1}\,\text{K11} + \frac{(\text{g2}\,\text{n2} - 1)}{\text{n2}}\,\text{g2}\,\text{K22} + \frac{(\text{g2bar}\,\text{n2bar} - 1)}{\text{n2bar}}\,\text{g2bar}\,\text{K2bar2bar} +\right.$$

$$\left.2\,\text{g1}\,\text{g2}\,\text{K12} + \frac{4}{3}\,\text{g2bar}\,\text{g1}\,\text{K2bar1} + \frac{4}{3}\,\text{g2bar}\,\text{g2}\,\text{K2bar2}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 +
  n2 * g2 * 1553 + n2bar * g2bar * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normalization for heavy anti-charm quark=  ", h20];
Print["normalization for heavy charm quark=  ", h30];
Print["normalization for light quark=  ", h40];
Print["normx2bar (computationally)=  ", normx2barr];
Print["normx2bar (by hand)=  ", normx2bar];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
Print["normx1 (computationally)=  ", normmx1];
Print["normx1 (by hand)=  ", normx1];
Print["0 to x2bar (h4 computationally)=  ", h4];
Print["0 to x2bar (h4 by hand)=  ", AB];
```

```
      Print["x2bar to x2 (h5 computationally)=  ", h5];
      Print["x2bar to x2 (h5 by hand)=  ", CD];
      Print["x2 to x1 (h6 computationally)=  ", h6];
      Print["x2 to x1 (h6 by hand)=  ", EF];
      Print["K2bar2bar  (computationally) = ", K2bar2barr];
      Print["K2bar2bar  (by hand) = ", K2bar2bar];
      Print["K2bar2  (computationally) = ", K2barr2];
      Print["K2bar2  (by hand) = ", K2bar2];
      Print["K2bar1  (computationally) = ", K2barr1];
      Print["K2bar1  (by hand) =", K2bar1];
      Print["K12  (computationally) = ", K122];
      Print["K12  (by hand) = ", K12];
      Print["K22  (computationally) = ", K222];
      Print["K22  (by hand) = ", K22];
      Print["K11  (computationally) = ", K111];
      Print["K11  (by hand) = ", K11];
      Print["x2= ", x2];
      Print["x1=  ", x1];
       Print["T by hand=", T1];
      Print["V by hand=", V1];
      Print["m2=", m2];
      Print["Energy=  ", Energy];
      Print["Energy without mass = ", Energywithoutmass];
      If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
        Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
         H = 10^9 + 1;
         m2 = m2B;
         If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
     If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
     If[n == 3, Energy6bare = HX;
      Print["x1=", x1];
      Print["x2=", x2];
      Print["x2bar=", x2bar];
      Print["Kinetic Energy=", T1];
      Print["Potential Energy=", V1];
      Print["Volume Energy=", Volumeenergy];
     ]];

(*Code for Plotting function *)
Plot1 = Plot[Evaluate[((y2bar[x] /.solution2bar) / x)^(3 / 2)],
    {x, .00001, x2bar}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
    (*Solution 1 = f3(x3), solution2 = f2(x2), solution3 = f1(x1) *)
```

```
    PlotRange → All,
    AxesLabel → {StyleForm["x", Italic, FontSize → 17]
      , StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]},
    PlotStyle -> {Thickness[0.01], Orange}, AxesStyle → Black,
    PlotLabel → None, LabelStyle → {GrayLevel[0], Bold}, ImageSize → Large];
(*data1=Cases[Plot[Evaluate[((y2bar[x]/.solution2bar)/x)^(3/2)],{x,.00001,x2bar}],
    Line[data1_]:→data1,∞][[1]];
Export["file1.txt",data1,"Table"];*)
Plot2 = Plot[Evaluate[((β22bar * y2bar[x] /.solution2bar) / x)^(3 / 2)], {x, .00001, x2bar},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel →
   {StyleForm["x", Italic, FontSize → 17], StyleForm["(f2(x)/ x)^(3/2)", Italic, FontSize → 17]},
  PlotStyle -> {Thickness[0.01], Green}, AxesStyle → Black, PlotLabel → None,
  LabelStyle → {GrayLevel[0], Bold}, ImageSize → Large]
Plot3 = Plot[Evaluate[((y2[x] /.solution2) / x)^(3 / 2)],
  {x, x2bar, x2}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel →
   {StyleForm["x", Italic, FontSize → 17], StyleForm["(f2(x)/ x)^(3/2)", Italic, FontSize → 17]},
  PlotStyle -> {Thickness[0.01], Green}, AxesStyle → Black, PlotLabel → None,
  LabelStyle → {GrayLevel[0], Bold}, ImageSize → Large]
Plot4 = Plot[Evaluate[((β12bar * y2bar[x] /.solution2bar) / x)^(3 / 2)], {x, .00001, x2bar},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel →
   {StyleForm["x", Italic, FontSize → 17], StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]},
  PlotStyle -> {Thickness[0.01], Blue}, AxesStyle → Black, PlotLabel → None,
  LabelStyle → {GrayLevel[0], Bold}, ImageSize → Large]
Plot5 = Plot[Evaluate[((γ12 * y2[x] /. solution2) / x)^(3 / 2)],
  {x, x2bar, x2}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel →
   {StyleForm["x", Italic, FontSize → 17], StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]},
  PlotStyle -> {Thickness[0.01], Blue}, AxesStyle → Black, PlotLabel → None,
  LabelStyle → {GrayLevel[0], Bold}, ImageSize → Large]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)],
  {x, x2, x1}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel →
   {StyleForm["x", Italic, FontSize → 17], StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]},
  PlotStyle -> {Thickness[0.01], Blue}, AxesStyle → Black, PlotLabel → None,
  LabelStyle → {GrayLevel[0], Bold}, ImageSize → Large]
```

```
Show[{Plot1}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
Show[Plot1, Plot2, Plot3, Plot4, Plot5,
 Plot6, AxesLabel → {StyleForm["x", Italic, FontSize → 17]
   , StyleForm["(f(x)/x)^3/2", Italic, FontSize → 17]}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

```
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333332
normalization for heavy charm quark=  0.333333389
normalization for light quark=  0.333333297
normx2bar (computationally)=  -0.699049095
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562429
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509031
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  2.78745825
0 to x2bar (h4 by hand)=  2.78745827
x2bar to x2 (h5 computationally)=  0.142631029
x2bar to x2 (h5 by hand)=  0.142631238
x2 to x1 (h6 computationally)=  0.775617917
x2 to x1 (h6 by hand)=  0.775617869
K2bar2bar  (computationally) = 0.389622494
K2bar2bar  (by hand) = 0.3896225
K2bar2  (computationally) = 0.385580345
K2bar2  (by hand) = 0.385580351
K2bar1  (computationally) = 0.251834613
K2bar1  (by hand) =0.251834614
```

```
K12  (computationally) = 0.251249946
K12  (by hand) = 0.251250008
K22  (computationally) = 0.381762331
K22  (by hand) = 0.381762581
K11  (computationally) = 0.216868789
K11  (by hand) = 0.216868699
x2= 0.353347815
x1=  0.610574059
T by hand=109.021747
V by hand=-46.863775
m2=9
Energy=  4212.26411
Energy without mass = 188.264112
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333332
normalization for heavy charm quark=  0.333333397
normalization for light quark=  0.333333293
normx2bar (computationally)=  -0.699049094
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562436
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509031
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  2.79067524
0 to x2bar (h4 by hand)=  2.79067525
x2bar to x2 (h5 computationally)=  0.142800731
x2bar to x2 (h5 by hand)=  0.142800974
x2 to x1 (h6 computationally)=  0.776585821
x2 to x1 (h6 by hand)=  0.776585763
K2bar2bar  (computationally) = 0.389844887
K2bar2bar  (by hand) = 0.389844893
K2bar2  (computationally) = 0.385800573
K2bar2  (by hand) = 0.385800578
K2bar1  (computationally) = 0.251983748
K2bar1  (by hand) =0.251983749
K12  (computationally) = 0.251398782
```

```
K12  (by hand) = 0.251398854
K22  (computationally) = 0.381980569
K22  (by hand) = 0.38198086
K11  (computationally) = 0.216999371
K11  (by hand) = 0.216999261
x2= 0.353143439
x1=  0.610198177
T by hand=109.153941
V by hand=-46.8915527
m2=9.01
Energy=  4212.13577
Energy without mass = 188.135771
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333381
normalization for light quark=  0.333333303
normx2bar (computationally)=  -0.699049096
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562423
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509032
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  2.819631
0 to x2bar (h4 by hand)=  2.81963101
x2bar to x2 (h5 computationally)=  0.144327597
x2bar to x2 (h5 by hand)=  0.144327774
x2 to x1 (h6 computationally)=  0.78529959
x2 to x1 (h6 by hand)=  0.785299553
K2bar2bar  (computationally) = 0.39184091
K2bar2bar  (by hand) = 0.391840917
K2bar2  (computationally) = 0.387777063
K2bar2  (by hand) = 0.38777707
K2bar1  (computationally) = 0.253322243
K2bar1  (by hand) =0.253322244
K12  (computationally) = 0.252734424
K12  (by hand) = 0.252734477
```

```
K22  (computationally) = 0.383938499
K22  (by hand) = 0.383938713
K11  (computationally) = 0.218171623
K11  (by hand) = 0.218171554
x2= 0.351319649
x1=  0.606845445
T by hand=110.34395
V by hand=-47.1408719
m2=9.1
Energy=  4211.01301
Energy without mass = 187.013007
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

```
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333394
normalization for light quark=  0.333333285
normx2bar (computationally)=  -0.699049097
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562434
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  2.85181008
0 to x2bar (h4 by hand)=  2.8518101
x2bar to x2 (h5 computationally)=  0.146024911
x2bar to x2 (h5 by hand)=  0.146025143
x2 to x1 (h6 computationally)=  0.794986548
x2 to x1 (h6 by hand)=  0.794986478
K2bar2bar  (computationally) = 0.39404714
K2bar2bar  (by hand) = 0.394047148
K2bar2  (computationally) = 0.389961733
K2bar2  (by hand) = 0.38996174
K2bar1  (computationally) = 0.254801611
K2bar1  (by hand) =0.254801612
```

```
K12  (computationally) = 0.254210696
K12  (by hand) = 0.254210765
K22  (computationally) = 0.386102896
K22  (by hand) = 0.386103175
K11  (computationally) = 0.219467057
K11  (by hand) = 0.219466919
x2= 0.349325615
x1=  0.603182634
T by hand=111.666709
V by hand=-47.4164195
m2=9.2
Energy=  4209.83184
Energy without mass = 185.831839
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333396
normalization for light quark=  0.333333295
normx2bar (computationally)=  -0.699049097
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562437
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509033
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  2.88399541
0 to x2bar (h4 by hand)=  2.88399543
x2bar to x2 (h5 computationally)=  0.147722716
x2bar to x2 (h5 by hand)=  0.147722962
x2 to x1 (h6 computationally)=  0.804678865
x2 to x1 (h6 by hand)=  0.804678807
K2bar2bar  (computationally) = 0.396241387
K2bar2bar  (by hand) = 0.396241395
K2bar2  (computationally) = 0.392134526
K2bar2  (by hand) = 0.392134533
K2bar1  (computationally) = 0.256272879
K2bar1  (by hand) =0.25627288
K12  (computationally) = 0.255678866
```

K12 (by hand) = 0.255678939

K22 (computationally) = 0.388255444

K22 (by hand) = 0.388255739

K11 (computationally) = 0.22075551

K11 (by hand) = 0.2207554

x2= 0.347364761

x1= 0.599583761

T by hand=112.990032

V by hand=-47.6904713

m2=9.3

Energy= 4208.71783

Energy without mass = 184.717829

R= 0.00995545615

normalization for heavy anti-charm quark= 0.333333333

normalization for heavy charm quark= 0.333333389

normalization for light quark= 0.333333294

normx2bar (computationally)= -0.699049098

normx2bar (by hand)= -0.699049152

normx2 (computationally)= -0.688562431

normx2 (by hand)= -0.68856244

normx1 (computationally)= -0.398509031

normx1 (by hand)= -0.398509033

0 to x2bar (h4 computationally)= 2.91618688

0 to x2bar (h4 by hand)= 2.91618689

x2bar to x2 (h5 computationally)= 0.149421004

x2bar to x2 (h5 by hand)= 0.149421221

x2 to x1 (h6 computationally)= 0.814376355

x2 to x1 (h6 by hand)= 0.814376299

K2bar2bar (computationally) = 0.398423842

K2bar2bar (by hand) = 0.398423851

K2bar2 (computationally) = 0.394295631

K2bar2 (by hand) = 0.394295639

K2bar1 (computationally) = 0.25773616

K2bar1 (by hand) =0.257736161

K12 (computationally) = 0.257139047

K12 (by hand) = 0.257139111

```
K22  (computationally) = 0.390396333
K22  (by hand) = 0.390396595
K11  (computationally) = 0.222036886
K11  (by hand) = 0.222036777
x2= 0.345436181
x1=  0.59604698
T by hand=114.313896
V by hand=-47.9630278
m2=9.4
Energy=  4207.66833
Energy without mass = 183.66833
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333325
normalization for heavy charm quark=  0.333333391
normalization for light quark=  0.3333333
normx2bar (computationally)=  -0.699049081
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562424
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.39850903
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  2.94838432
0 to x2bar (h4 by hand)=  2.94838431
x2bar to x2 (h5 computationally)=  0.151119987
x2bar to x2 (h5 by hand)=  0.15112025
x2 to x1 (h6 computationally)=  0.82407897
x2 to x1 (h6 by hand)=  0.824078922
K2bar2bar  (computationally) = 0.400594654
K2bar2bar  (by hand) = 0.400594646
K2bar2  (computationally) = 0.396445225
K2bar2  (by hand) = 0.396445217
K2bar1  (computationally) = 0.259191588
K2bar1  (by hand) =0.259191587
K12  (computationally) = 0.258591424
K12  (by hand) = 0.2585915
K22  (computationally) = 0.392525958
```

K22 (by hand) = 0.39252626

K11 (computationally) = 0.223311386

K11 (by hand) = 0.223311298

x2= 0.343539009

x1= 0.592570553

T by hand=115.638299

V by hand=-48.2341272

m2=9.5

Energy= 4206.68083

Energy without mass = 182.680831

R= 0.00995545615

normalization for heavy anti-charm quark= 0.333333327

normalization for heavy charm quark= 0.333333388

normalization for light quark= 0.333333295

normx2bar (computationally)= -0.699049085

normx2bar (by hand)= -0.699049152

normx2 (computationally)= -0.688562423

normx2 (by hand)= -0.68856244

normx1 (computationally)= -0.398509028

normx1 (by hand)= -0.398509033

0 to x2bar (h4 computationally)= 2.9805878

0 to x2bar (h4 by hand)= 2.98058779

x2bar to x2 (h5 computationally)= 0.15281939

x2bar to x2 (h5 by hand)= 0.152819632

x2 to x1 (h6 computationally)= 0.833786589

x2 to x1 (h6 by hand)= 0.833786533

K2bar2bar (computationally) = 0.402754093

K2bar2bar (by hand) = 0.402754089

K2bar2 (computationally) = 0.398583538

K2bar2 (by hand) = 0.398583534

K2bar1 (computationally) = 0.260639287

K2bar1 (by hand) =0.260639286

K12 (computationally) = 0.260036058

K12 (by hand) = 0.260036128

K22 (computationally) = 0.394644282

K22 (by hand) = 0.394644563

```
K11  (computationally) = 0.224579026
K11  (by hand) = 0.22457892
x2= 0.3416724
x1=  0.589152786
T by hand=116.963227
V by hand=-48.5037776
m2=9.6
Energy=  4205.75295
Energy without mass = 181.752948
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333385
normalization for light quark=  0.333333299
normx2bar (computationally)=  -0.699049098
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562428
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509033
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.01279716
0 to x2bar (h4 by hand)=  3.01279718
x2bar to x2 (h5 computationally)=  0.154519307
x2bar to x2 (h5 by hand)=  0.154519517
x2 to x1 (h6 computationally)=  0.843499191
x2 to x1 (h6 by hand)=  0.843499141
K2bar2bar  (computationally) = 0.404902312
K2bar2bar  (by hand) = 0.404902321
K2bar2  (computationally) = 0.400710737
K2bar2  (by hand) = 0.400710745
K2bar1  (computationally) = 0.262079388
K2bar1  (by hand) =0.262079389
K12  (computationally) = 0.261473105
K12  (by hand) = 0.261473167
K22  (computationally) = 0.396751572
K22  (by hand) = 0.396751826
K11  (computationally) = 0.225840038
```

K11 (by hand) = 0.225839943

x2= 0.33983555

x1= 0.585792075

T by hand=118.288678

V by hand=−48.7720177

m2=9.7

Energy= 4204.88241

Energy without mass = 180.882414

R= 0.00995545615

normalization for heavy anti-charm quark= 0.333333316

normalization for heavy charm quark= 0.333333405

normalization for light quark= 0.333333296

normx2bar (computationally)= −0.699049062

normx2bar (by hand)= −0.699049152

normx2 (computationally)= −0.688562424

normx2 (by hand)= −0.68856244

normx1 (computationally)= −0.398509027

normx1 (by hand)= −0.398509033

0 to x2bar (h4 computationally)= 3.0450121

0 to x2bar (h4 by hand)= 3.04501203

x2bar to x2 (h5 computationally)= 0.156220116

x2bar to x2 (h5 by hand)= 0.156220489

x2 to x1 (h6 computationally)= 0.853216632

x2 to x1 (h6 by hand)= 0.853216568

K2bar2bar (computationally) = 0.407039344

K2bar2bar (by hand) = 0.407039315

K2bar2 (computationally) = 0.402826905

K2bar2 (by hand) = 0.402826878

K2bar1 (computationally) = 0.263511985

K2bar1 (by hand) =0.263511981

K12 (computationally) = 0.262902751

K12 (by hand) = 0.262902856

K22 (computationally) = 0.398848299

K22 (by hand) = 0.398848711

K11 (computationally) = 0.227094394

K11 (by hand) = 0.227094279

```
x2= 0.338027685
x1=  0.582486852
T by hand=119.614641
V by hand=-49.0388612
m2=9.8
Energy=  4204.06707
Energy without mass = 180.067068
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333316
normalization for heavy charm quark=  0.333333414
normalization for light quark=  0.333333301
normx2bar (computationally)=  -0.699049062
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562432
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509032
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.07723285
0 to x2bar (h4 by hand)=  3.07723279
x2bar to x2 (h5 computationally)=  0.157921231
x2bar to x2 (h5 by hand)=  0.157921652
x2 to x1 (h6 computationally)=  0.862938913
x2 to x1 (h6 by hand)=  0.862938852
K2bar2bar  (computationally) = 0.409165564
K2bar2bar  (by hand) = 0.409165534
K2bar2  (computationally) = 0.404932337
K2bar2  (by hand) = 0.404932309
K2bar1  (computationally) = 0.264937226
K2bar1  (by hand) =0.264937221
K12  (computationally) = 0.264325004
K12  (by hand) = 0.264325123
K22  (computationally) = 0.400934173
K22  (by hand) = 0.400934641
K11  (computationally) = 0.228342333
K11  (by hand) = 0.228342226
x2= 0.336248045
```

x1= 0.579235632

T by hand=120.941111

V by hand=−49.3043352

m2=9.9

Energy= 4203.30486

Energy without mass = 179.304862

R= 0.00995545615

normalization for heavy anti-charm quark= 0.333333313

normalization for heavy charm quark= 0.333333414

normalization for light quark= 0.333333294

normx2bar (computationally)= −0.699049056

normx2bar (by hand)= −0.699049152

normx2 (computationally)= −0.688562428

normx2 (by hand)= −0.68856244

normx1 (computationally)= −0.398509027

normx1 (by hand)= −0.398509033

0 to x2bar (h4 computationally)= 3.10945918

0 to x2bar (h4 by hand)= 3.1094591

x2bar to x2 (h5 computationally)= 0.159622801

x2bar to x2 (h5 by hand)= 0.159623239

x2 to x1 (h6 computationally)= 0.872665901

x2 to x1 (h6 by hand)= 0.872665827

K2bar2bar (computationally) = 0.411281037

K2bar2bar (by hand) = 0.411281

K2bar2 (computationally) = 0.40702712

K2bar2 (by hand) = 0.407027085

K2bar1 (computationally) = 0.266355199

K2bar1 (by hand) =0.266355194

K12 (computationally) = 0.265739991

K12 (by hand) = 0.265740114

K22 (computationally) = 0.403009439

K22 (by hand) = 0.403009921

K11 (computationally) = 0.229583817

K11 (by hand) = 0.229583683

x2= 0.334495904

x1= 0.57603696

```
T by hand=122.268072
V by hand=-49.568447
m2=10.
Energy=  4202.59384
Energy without mass = 178.59384
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333318
normalization for heavy charm quark=  0.3333334
normalization for light quark=  0.333333302
normx2bar (computationally)=  -0.699049066
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562422
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.39850903
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.14169109
0 to x2bar (h4 by hand)=  3.14169103
x2bar to x2 (h5 computationally)=  0.161324693
x2bar to x2 (h5 by hand)=  0.161325049
x2 to x1 (h6 computationally)=  0.882397614
x2 to x1 (h6 by hand)=  0.882397561
K2bar2bar  (computationally) = 0.413385977
K2bar2bar  (by hand) = 0.413385952
K2bar2  (computationally) = 0.409111449
K2bar2  (by hand) = 0.409111424
K2bar1  (computationally) = 0.267766039
K2bar1  (by hand) =0.267766035
K12  (computationally) = 0.267147815
K12  (by hand) = 0.267147915
K22  (computationally) = 0.405074162
K22  (by hand) = 0.405074557
K11  (computationally) = 0.230819129
K11  (by hand) = 0.230819035
x2= 0.332770565
x1=  0.572889463
T by hand=123.595529
```

```
V by hand=-49.831236
m2=10.1
Energy=  4201.93214
Energy without mass = 177.93214
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333392
normalization for light quark=  0.333333285
normx2bar (computationally)=  -0.699049098
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562433
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509029
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.17392851
0 to x2bar (h4 by hand)=  3.17392853
x2bar to x2 (h5 computationally)=  0.163027047
x2bar to x2 (h5 by hand)=  0.163027301
x2 to x1 (h6 computationally)=  0.892133844
x2 to x1 (h6 by hand)=  0.892133764
K2bar2bar  (computationally) = 0.415480557
K2bar2bar  (by hand) = 0.415480566
K2bar2  (computationally) = 0.41118551
K2bar2  (by hand) = 0.411185519
K2bar1  (computationally) = 0.269169828
K2bar1  (by hand) =0.269169829
K12  (computationally) = 0.268548595
K12  (by hand) = 0.26854867
K22  (computationally) = 0.407128677
K22  (by hand) = 0.407128983
K11  (computationally) = 0.232048092
K11  (by hand) = 0.232047934
x2= 0.331071359
x1=  0.56979178
T by hand=124.923459
V by hand=-50.0926956
```

```
m2=10.2
Energy=  4201.31799
Energy without mass = 177.317987
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333312
normalization for heavy charm quark=  0.333333409
normalization for light quark=  0.333333301
normx2bar (computationally)=  -0.699049053
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562422
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509029
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.20617095
0 to x2bar (h4 by hand)=  3.20617087
x2bar to x2 (h5 computationally)=  0.164730413
x2bar to x2 (h5 by hand)=  0.16473085
x2 to x1 (h6 computationally)=  0.901874677
x2 to x1 (h6 by hand)=  0.901874615
K2bar2bar  (computationally) = 0.417564678
K2bar2bar  (by hand) = 0.417564636
K2bar2  (computationally) = 0.413249285
K2bar2  (by hand) = 0.413249246
K2bar1  (computationally) = 0.270566679
K2bar1  (by hand) =0.270566673
K12  (computationally) = 0.269942576
K12  (by hand) = 0.269942698
K22  (computationally) = 0.409173529
K22  (by hand) = 0.409174005
K11  (computationally) = 0.233271124
K11  (by hand) = 0.233271018
x2= 0.329397645
x1=  0.566742653
T by hand=126.251879
V by hand=-50.3528902
m2=10.3
```

```
Energy=  4200.74968
Energy without mass = 176.749681
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333328
normalization for heavy charm quark=  0.333333393
normalization for light quark=  0.333333289
normx2bar (computationally)=  -0.699049087
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562428
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509028
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.23841901
0 to x2bar (h4 by hand)=  3.23841901
x2bar to x2 (h5 computationally)=  0.166433661
x2bar to x2 (h5 by hand)=  0.166433951
x2 to x1 (h6 computationally)=  0.911619925
x2 to x1 (h6 by hand)=  0.91161985
K2bar2bar  (computationally) = 0.419638889
K2bar2bar  (by hand) = 0.419638887
K2bar2  (computationally) = 0.415303167
K2bar2  (by hand) = 0.415303164
K2bar1  (computationally) = 0.271956701
K2bar1  (by hand) =0.2719567
K12  (computationally) = 0.271329599
K12  (by hand) = 0.271329683
K22  (computationally) = 0.411207969
K22  (by hand) = 0.411208308
K11  (computationally) = 0.234488018
K11  (by hand) = 0.234487873
x2= 0.327748772
x1=  0.563740808
T by hand=127.580755
V by hand=-50.6117865
m2=10.4
Energy=  4200.22562
```

```
Energy without mass = 176.225617
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333318
normalization for heavy charm quark=  0.333333414
normalization for light quark=  0.333333287
normx2bar (computationally)=  -0.699049065
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562433
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.27067202
0 to x2bar (h4 by hand)=  3.27067196
x2bar to x2 (h5 computationally)=  0.168137935
x2bar to x2 (h5 by hand)=  0.168138381
x2 to x1 (h6 computationally)=  0.921369567
x2 to x1 (h6 by hand)=  0.921369476
K2bar2bar  (computationally) = 0.421702987
K2bar2bar  (by hand) = 0.421702959
K2bar2  (computationally) = 0.417347103
K2bar2  (by hand) = 0.417347076
K2bar1  (computationally) = 0.273339978
K2bar1  (by hand) =0.273339974
K12  (computationally) = 0.272710022
K12  (by hand) = 0.272710149
K22  (computationally) = 0.413233115
K22  (by hand) = 0.413233614
K11  (computationally) = 0.235699009
K11  (by hand) = 0.235698839
x2= 0.32612416
x1=  0.560785061
T by hand=128.910099
V by hand=-50.8694402
m2=10.5
Energy=  4199.74424
Energy without mass = 175.744242
```

```
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333328
normalization for heavy charm quark=  0.333333393
normalization for light quark=  0.333333297
normx2bar (computationally)=  -0.699049087
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562428
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509032
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.30293034
0 to x2bar (h4 by hand)=  3.30293034
x2bar to x2 (h5 computationally)=  0.169842127
x2bar to x2 (h5 by hand)=  0.169842426
x2 to x1 (h6 computationally)=  0.931123629
x2 to x1 (h6 by hand)=  0.931123566
K2bar2bar  (computationally) = 0.423757376
K2bar2bar  (by hand) = 0.423757373
K2bar2  (computationally) = 0.419381357
K2bar2  (by hand) = 0.419381354
K2bar1  (computationally) = 0.274716637
K2bar1  (by hand) =0.274716637
K12  (computationally) = 0.274083709
K12  (by hand) = 0.274083796
K22  (computationally) = 0.415248107
K22  (by hand) = 0.415248456
K11  (computationally) = 0.236904291
K11  (by hand) = 0.236904172
x2= 0.324523195
x1=  0.55787426
T by hand=130.239901
V by hand=-51.1258601
m2=10.6
Energy=  4199.30409
Energy without mass = 175.304092
R=  0.00995545615
```

```
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333392
normalization for light quark=  0.333333281
normx2bar (computationally)=  -0.699049098
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562433
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509028
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.33519367
0 to x2bar (h4 by hand)=  3.33519369
x2bar to x2 (h5 computationally)=  0.171546997
x2bar to x2 (h5 by hand)=  0.171547267
x2 to x1 (h6 computationally)=  0.940881871
x2 to x1 (h6 by hand)=  0.940881779
K2bar2bar  (computationally) = 0.425802051
K2bar2bar  (by hand) = 0.425802061
K2bar2  (computationally) = 0.421406016
K2bar2  (by hand) = 0.421406025
K2bar1  (computationally) = 0.276086735
K2bar1  (by hand) =0.276086736
K12  (computationally) = 0.2754509
K12  (by hand) = 0.27545098
K22  (computationally) = 0.417253812
K22  (by hand) = 0.417254138
K11  (computationally) = 0.238103633
K11  (by hand) = 0.238103449
x2= 0.322945331
x1=  0.555007264
T by hand=131.570143
V by hand=-51.3810444
m2=10.7
Energy=  4198.90375
Energy without mass = 174.903755
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333327
normalization for heavy charm quark=  0.333333401
normalization for light quark=  0.333333281
normx2bar (computationally)=  -0.699049085
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562434
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.36746187
0 to x2bar (h4 by hand)=  3.36746186
x2bar to x2 (h5 computationally)=  0.173252519
x2bar to x2 (h5 by hand)=  0.173252869
x2 to x1 (h6 computationally)=  0.950644371
x2 to x1 (h6 by hand)=  0.950644272
K2bar2bar  (computationally) = 0.427837113
K2bar2bar  (by hand) = 0.427837108
K2bar2  (computationally) = 0.423421181
K2bar2  (by hand) = 0.423421176
K2bar1  (computationally) = 0.277450385
K2bar1  (by hand) =0.277450384
K12  (computationally) = 0.276811703
K12  (by hand) = 0.276811805
K22  (computationally) = 0.419250311
K22  (by hand) = 0.419250717
K11  (computationally) = 0.239297394
K11  (by hand) = 0.239297201
x2= 0.321390018
x1=  0.552183016
T by hand=132.900835
V by hand=-51.6350402
m2=10.8
Energy=  4198.54188
Energy without mass = 174.541882
```

```
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333311
normalization for heavy charm quark=  0.333333422
normalization for light quark=  0.33333329
normx2bar (computationally)=  -0.699049051
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562432
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.39973487
0 to x2bar (h4 by hand)=  3.39973477
x2bar to x2 (h5 computationally)=  0.174958695
x2bar to x2 (h5 by hand)=  0.174959235
x2 to x1 (h6 computationally)=  0.960411047
x2 to x1 (h6 by hand)=  0.96041095
K2bar2bar  (computationally) = 0.429862697
K2bar2bar  (by hand) = 0.429862651
K2bar2  (computationally) = 0.425426985
K2bar2  (by hand) = 0.425426941
K2bar1  (computationally) = 0.278807673
K2bar1  (by hand) =0.278807666
K12  (computationally) = 0.278166208
K12  (by hand) = 0.278166359
K22  (computationally) = 0.421237743
K22  (by hand) = 0.421238336
K11  (computationally) = 0.240485604
K11  (by hand) = 0.240485428
x2= 0.319856727
x1=  0.549400471
T by hand=134.231968
V by hand=-51.8878592
m2=10.9
Energy=  4198.21719
Energy without mass = 174.217187
R=  0.00995545615
```

```
normalization for heavy anti-charm quark=  0.333333311
normalization for heavy charm quark=  0.333333411
normalization for light quark=  0.333333295
normx2bar (computationally)=  -0.699049051
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562422
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.43201287
0 to x2bar (h4 by hand)=  3.43201277
x2bar to x2 (h5 computationally)=  0.176664877
x2bar to x2 (h5 by hand)=  0.176665368
x2 to x1 (h6 computationally)=  0.970181854
x2 to x1 (h6 by hand)=  0.970181773
K2bar2bar  (computationally) = 0.431879085
K2bar2bar  (by hand) = 0.431879037
K2bar2  (computationally) = 0.427423629
K2bar2  (by hand) = 0.427423584
K2bar1  (computationally) = 0.280158698
K2bar1  (by hand) =0.280158691
K12  (computationally) = 0.279514356
K12  (by hand) = 0.279514493
K22  (computationally) = 0.423215654
K22  (by hand) = 0.423216188
K11  (computationally) = 0.241668322
K11  (by hand) = 0.241668177
x2= 0.31834493
x1=  0.546658615
T by hand=135.56353
V by hand=-52.1395029
m2=11.
Energy=  4197.92844
Energy without mass = 173.928444
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
```

```
normalization for heavy charm quark=  0.333333395
normalization for light quark=  0.333333278
normx2bar (computationally)=  -0.699049097
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562436
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.46429587
0 to x2bar (h4 by hand)=  3.46429589
x2bar to x2 (h5 computationally)=  0.178371249
x2bar to x2 (h5 by hand)=  0.178371552
x2 to x1 (h6 computationally)=  0.979956648
x2 to x1 (h6 by hand)=  0.979956543
K2bar2bar  (computationally) = 0.43388644
K2bar2bar  (by hand) = 0.433886449
K2bar2  (computationally) = 0.429411294
K2bar2  (by hand) = 0.429411303
K2bar1  (computationally) = 0.281503538
K2bar1  (by hand) =0.28150354
K12  (computationally) = 0.280856272
K12  (by hand) = 0.280856362
K22  (computationally) = 0.425184415
K22  (by hand) = 0.425184779
K11  (computationally) = 0.242845472
K11  (by hand) = 0.242845261
x2= 0.31685414
x1=  0.543956466
T by hand=136.895507
V by hand=-52.3899758
m2=11.1
Energy=  4197.67447
Energy without mass = 173.674473
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333311
normalization for heavy charm quark=  0.33333341
```

```
normalization for light quark=  0.333333305
normx2bar (computationally)=  −0.69904905
normx2bar (by hand)=  −0.699049152
normx2 (computationally)=  −0.688562421
normx2 (by hand)=  −0.68856244
normx1 (computationally)=  −0.398509031
normx1 (by hand)=  −0.398509033
0 to x2bar (h4 computationally)=  3.49658315
0 to x2bar (h4 by hand)=  3.49658304
x2bar to x2 (h5 computationally)=  0.180078744
x2bar to x2 (h5 by hand)=  0.180079242
x2 to x1 (h6 computationally)=  0.989735547
x2 to x1 (h6 by hand)=  0.989735484
K2bar2bar  (computationally) = 0.435884499
K2bar2bar  (by hand) = 0.43588445
K2bar2  (computationally) = 0.431389843
K2bar2  (by hand) = 0.431389796
K2bar1  (computationally) = 0.282842276
K2bar1  (by hand) =0.282842269
K12  (computationally) = 0.282192266
K12  (by hand) = 0.282192405
K22  (computationally) = 0.42714485
K22  (by hand) = 0.427145391
K11  (computationally) = 0.24401751
K11  (by hand) = 0.244017404
x2= 0.31538389
x1=  0.541293113
T by hand=138.227924
V by hand=−52.6393516
m2=11.2
Energy=  4197.45413
Energy without mass = 173.454134
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333311
normalization for heavy charm quark=  0.33333341
normalization for light quark=  0.333333296
```

```
normx2bar (computationally)=  -0.699049049
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562421
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.52887529
0 to x2bar (h4 by hand)=  3.52887519
x2bar to x2 (h5 computationally)=  0.181786324
x2bar to x2 (h5 by hand)=  0.18178683
x2 to x1 (h6 computationally)=  0.999518287
x2 to x1 (h6 by hand)=  0.999518204
K2bar2bar  (computationally) = 0.437873771
K2bar2bar  (by hand) = 0.437873721
K2bar2  (computationally) = 0.433359645
K2bar2  (by hand) = 0.433359598
K2bar1  (computationally) = 0.284174986
K2bar1  (by hand) =0.284174979
K12  (computationally) = 0.283522163
K12  (by hand) = 0.283522304
K22  (computationally) = 0.429096273
K22  (by hand) = 0.429096821
K11  (computationally) = 0.245184031
K11  (by hand) = 0.245183884
x2= 0.313933699
x1=  0.538667605
T by hand=139.560738
V by hand=-52.8875746
m2=11.3
Energy=  4197.26636
Energy without mass = 173.266358
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333317
normalization for heavy charm quark=  0.333333416
normalization for light quark=  0.333333289
normx2bar (computationally)=  -0.699049063
```

```
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562434
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509029
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.56117209
0 to x2bar (h4 by hand)=  3.56117202
x2bar to x2 (h5 computationally)=  0.183494325
x2bar to x2 (h5 by hand)=  0.183494837
x2 to x1 (h6 computationally)=  1.00930492
x2 to x1 (h6 by hand)=  1.00930482
K2bar2bar  (computationally) = 0.439854283
K2bar2bar  (by hand) = 0.43985425
K2bar2  (computationally) = 0.435320771
K2bar2  (by hand) = 0.435320739
K2bar1  (computationally) = 0.285501765
K2bar1  (by hand) =0.28550176
K12  (computationally) = 0.28484614
K12  (by hand) = 0.284846284
K22  (computationally) = 0.431039097
K22  (by hand) = 0.431039668
K11  (computationally) = 0.246345356
K11  (by hand) = 0.246345168
x2= 0.312503124
x1=  0.536079083
T by hand=140.893963
V by hand=-53.1346943
m2=11.4
Energy=  4197.1101
Energy without mass = 173.110103
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333318
normalization for heavy charm quark=  0.333333416
normalization for light quark=  0.333333285
normx2bar (computationally)=  -0.699049065
normx2bar (by hand)=  -0.699049152
```

```
normx2 (computationally)=  -0.688562435
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509028
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.59347336
0 to x2bar (h4 by hand)=  3.59347329
x2bar to x2 (h5 computationally)=  0.18520273
x2bar to x2 (h5 by hand)=  0.185203242
x2 to x1 (h6 computationally)=  1.01909542
x2 to x1 (h6 by hand)=  1.01909531
K2bar2bar  (computationally) = 0.441826081
K2bar2bar  (by hand) = 0.44182605
K2bar2  (computationally) = 0.437273267
K2bar2  (by hand) = 0.437273238
K2bar1  (computationally) = 0.286822682
K2bar1  (by hand) =0.286822677
K12  (computationally) = 0.286164265
K12  (by hand) = 0.28616441
K22  (computationally) = 0.432973362
K22  (by hand) = 0.432973934
K11  (computationally) = 0.247501553
K11  (by hand) = 0.247501347
x2= 0.311091724
x1=  0.533526692
T by hand=142.227591
V by hand=-53.3807237
m2=11.5
Energy=  4196.98437
Energy without mass = 172.984375
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333396
normalization for light quark=  0.333333275
normx2bar (computationally)=  -0.699049096
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562436
```

```
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509026
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.62577925
0 to x2bar (h4 by hand)=  3.62577927
x2bar to x2 (h5 computationally)=  0.186911207
x2bar to x2 (h5 by hand)=  0.186911531
x2 to x1 (h6 computationally)=  1.0288897
x2 to x1 (h6 by hand)=  1.02888959
K2bar2bar  (computationally) = 0.443789395
K2bar2bar  (by hand) = 0.443789404
K2bar2  (computationally) = 0.439217321
K2bar2  (by hand) = 0.43921733
K2bar1  (computationally) = 0.288137822
K2bar1  (by hand) =0.288137824
K12  (computationally) = 0.287476543
K12  (by hand) = 0.287476639
K22  (computationally) = 0.434898916
K22  (by hand) = 0.434899305
K11  (computationally) = 0.248652654
K11  (by hand) = 0.248652418
x2= 0.309699071
x1=  0.531009602
T by hand=143.561614
V by hand=-53.6256673
m2=11.6
Energy=  4196.88822
Energy without mass = 172.888223
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333392
normalization for light quark=  0.333333277
normx2bar (computationally)=  -0.699049096
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562433
normx2 (by hand)=  -0.68856244
```

```
normx1 (computationally)=  -0.398509026
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.65808939
0 to x2bar (h4 by hand)=  3.65808941
x2bar to x2 (h5 computationally)=  0.188620396
x2bar to x2 (h5 by hand)=  0.188620704
x2 to x1 (h6 computationally)=  1.03868776
x2 to x1 (h6 by hand)=  1.03868764
K2bar2bar  (computationally) = 0.44574416
K2bar2bar  (by hand) = 0.445744169
K2bar2  (computationally) = 0.441152947
K2bar2  (by hand) = 0.441152957
K2bar1  (computationally) = 0.289447253
K2bar1  (by hand) =0.289447254
K12  (computationally) = 0.288783198
K12  (by hand) = 0.288783289
K22  (computationally) = 0.436816436
K22  (by hand) = 0.436816805
K11  (computationally) = 0.249798799
K11  (by hand) = 0.249798569
x2= 0.308324769
x1=  0.528527022
T by hand=144.896036
V by hand=-53.8695597
m2=11.7
Energy=  4196.82073
Energy without mass = 172.820726
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333318
normalization for heavy charm quark=  0.333333402
normalization for light quark=  0.333333292
normx2bar (computationally)=  -0.699049064
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562422
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
```

```
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.69040373
0 to x2bar (h4 by hand)=  3.69040366
x2bar to x2 (h5 computationally)=  0.190330267
x2bar to x2 (h5 by hand)=  0.190330716
x2 to x1 (h6 computationally)=  1.04848954
x2 to x1 (h6 by hand)=  1.04848945
K2bar2bar  (computationally) = 0.447690479
K2bar2bar  (by hand) = 0.447690446
K2bar2  (computationally) = 0.443080245
K2bar2  (by hand) = 0.443080214
K2bar1  (computationally) = 0.290751049
K2bar1  (by hand) =0.290751044
K12  (computationally) = 0.290084299
K12  (by hand) = 0.290084425
K22  (computationally) = 0.438725994
K22  (by hand) = 0.438726491
K11  (computationally) = 0.250940074
K11  (by hand) = 0.250939905
x2= 0.306968418
x1=  0.526078178
T by hand=146.230853
V by hand=-54.1124163
m2=11.8
Energy=  4196.78101
Energy without mass = 172.781005
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333395
normalization for light quark=  0.333333278
normx2bar (computationally)=  -0.699049096
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562435
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509028
normx1 (by hand)=  -0.398509033
```

```
0 to x2bar (h4 computationally)=  3.72272269
0 to x2bar (h4 by hand)=  3.72272271
x2bar to x2 (h5 computationally)=  0.19204007
x2bar to x2 (h5 by hand)=  0.192040401
x2 to x1 (h6 computationally)=  1.05829489
x2 to x1 (h6 by hand)=  1.05829477
K2bar2bar  (computationally) = 0.449628745
K2bar2bar  (by hand) = 0.449628755
K2bar2  (computationally) = 0.444999508
K2bar2  (by hand) = 0.444999518
K2bar1  (computationally) = 0.292049286
K2bar1  (by hand) =0.292049288
K12  (computationally) = 0.291379741
K12  (by hand) = 0.291379839
K22  (computationally) = 0.440627117
K22  (by hand) = 0.440627514
K11  (computationally) = 0.252076291
K11  (by hand) = 0.25207606
x2= 0.305629622
x1=  0.5236623
T by hand=147.566039
V by hand=-54.35421
m2=11.9
Energy=  4196.76823
Energy without mass = 172.768227
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333324
normalization for heavy charm quark=  0.333333408
normalization for light quark=  0.333333281
normx2bar (computationally)=  -0.699049076
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562435
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509027
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.75504564
```

```
0 to x2bar (h4 by hand)=  3.7550456
x2bar to x2 (h5 computationally)=  0.193750703
x2bar to x2 (h5 by hand)=  0.193751164
x2 to x1 (h6 computationally)=  1.06810388
x2 to x1 (h6 by hand)=  1.06810376
K2bar2bar  (computationally) = 0.451558716
K2bar2bar  (by hand) = 0.4515587
K2bar2  (computationally) = 0.446910613
K2bar2  (by hand) = 0.446910597
K2bar1  (computationally) = 0.293342027
K2bar1  (by hand) =0.293342024
K12  (computationally) = 0.292669804
K12  (by hand) = 0.292669936
K22  (computationally) = 0.442520611
K22  (by hand) = 0.442521136
K11  (computationally) = 0.253207768
K11  (by hand) = 0.253207537
x2= 0.30430802
x1=  0.521278681
T by hand=148.901611
V by hand=-54.5949977
m2=12.
Energy=  4196.78157
Energy without mass = 172.781569
R=  0.00995545615
normalization for heavy anti-charm quark=  0.33333331
normalization for heavy charm quark=  0.333333422
normalization for light quark=  0.333333288
normx2bar (computationally)=  -0.699049047
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562429
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509026
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.72595456
0 to x2bar (h4 by hand)=  3.72595444
```

```
x2bar to x2 (h5 computationally)=  0.192211591
x2bar to x2 (h5 by hand)=  0.192212201
x2 to x1 (h6 computationally)=  1.05927563
x2 to x1 (h6 by hand)=  1.05927552
K2bar2bar  (computationally) = 0.449821968
K2bar2bar  (by hand) = 0.449821912
K2bar2  (computationally) = 0.4451909
K2bar2  (by hand) = 0.445190847
K2bar1  (computationally) = 0.292178797
K2bar1  (by hand) =0.292178789
K12  (computationally) = 0.291509099
K12  (by hand) = 0.29150927
K22  (computationally) = 0.440817234
K22  (by hand) = 0.4408179
K11  (computationally) = 0.252189686
K11  (by hand) = 0.252189481
x2= 0.305496703
x1=  0.523422503
T by hand=147.699585
V by hand=-54.3783469
m2=11.91
Energy=  4196.76839
Energy without mass = 172.768394
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333326
normalization for heavy charm quark=  0.333333407
normalization for light quark=  0.333333282
normx2bar (computationally)=  -0.69904908
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562436
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509028
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.72918688
0 to x2bar (h4 by hand)=  3.72918686
x2bar to x2 (h5 computationally)=  0.192382319
```

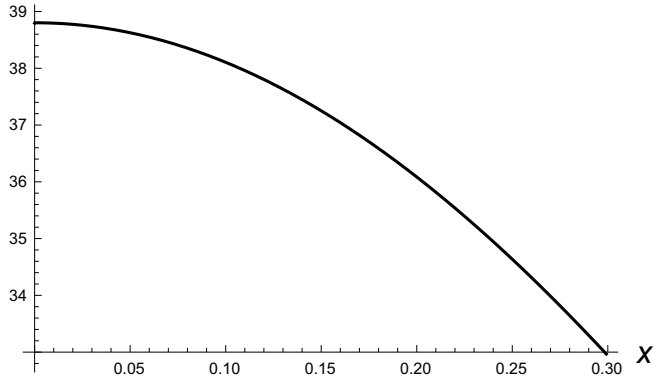
```
x2bar to x2 (h5 by hand)=  0.192382761
x2 to x1 (h6 computationally)=  1.0602564
x2 to x1 (h6 by hand)=  1.06025628
K2bar2bar  (computationally) = 0.450015351
K2bar2bar  (by hand) = 0.450015339
K2bar2  (computationally) = 0.445382353
K2bar2  (by hand) = 0.445382342
K2bar1  (computationally) = 0.292308269
K2bar1  (by hand) =0.292308267
K12  (computationally) = 0.291638225
K12  (by hand) = 0.291638352
K22  (computationally) = 0.441006591
K22  (by hand) = 0.4410071
K11  (computationally) = 0.252302979
K11  (by hand) = 0.252302754
x2= 0.305363941
x1=  0.52318302
T by hand=147.833125
V by hand=-54.402453
m2=11.92
Energy=  4196.76883
Energy without mass = 172.768829
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333332
normalization for heavy charm quark=  0.333333397
normalization for light quark=  0.333333275
normx2bar (computationally)=  -0.699049094
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.688562435
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509026
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.72627801
0 to x2bar (h4 by hand)=  3.72627802
x2bar to x2 (h5 computationally)=  0.192228224
x2bar to x2 (h5 by hand)=  0.192228571
```

```
x2 to x1 (h6 computationally)=  1.05937369
x2 to x1 (h6 by hand)=  1.05937356
K2bar2bar  (computationally) = 0.449841438
K2bar2bar  (by hand) = 0.449841445
K2bar2  (computationally) = 0.445210122
K2bar2  (by hand) = 0.445210129
K2bar1  (computationally) = 0.292191753
K2bar1  (by hand) =0.292191754
K12  (computationally) = 0.291521913
K12  (by hand) = 0.291522015
K22  (computationally) = 0.440835794
K22  (by hand) = 0.440836208
K11  (computationally) = 0.25220096
K11  (by hand) = 0.252200712
x2= 0.305483411
x1=  0.523398534
T by hand=147.712931
V by hand=-54.3807434
m2=11.911
Energy=  4196.76843
Energy without mass = 172.76843
R=  0.00995545615
normalization for heavy anti-charm quark=  0.333333333
normalization for heavy charm quark=  0.333333379
normalization for light quark=  0.333333283
normx2bar (computationally)=  -0.699049095
normx2bar (by hand)=  -0.699049152
normx2 (computationally)=  -0.68856242
normx2 (by hand)=  -0.68856244
normx1 (computationally)=  -0.398509026
normx1 (by hand)=  -0.398509033
0 to x2bar (h4 computationally)=  3.72660123
0 to x2bar (h4 by hand)=  3.72660125
x2bar to x2 (h5 computationally)=  0.192245137
x2bar to x2 (h5 by hand)=  0.192245379
x2 to x1 (h6 computationally)=  1.05947181
```

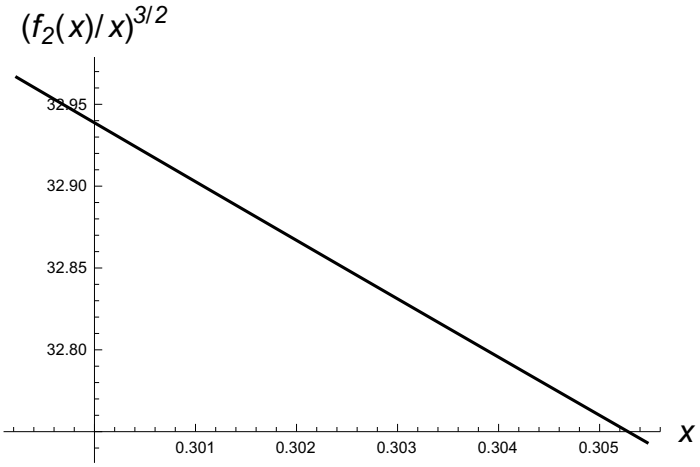
```
x2 to x1 (h6 by hand)=  1.05947172
K2bar2bar  (computationally) = 0.449860773
K2bar2bar  (by hand) = 0.449860781
K2bar2  (computationally) = 0.445229248
K2bar2  (by hand) = 0.445229256
K2bar1  (computationally) = 0.292204707
K2bar1  (by hand) =0.292204708
K12  (computationally) = 0.291534793
K12  (by hand) = 0.291534865
K22  (computationally) = 0.440854547
K22  (by hand) = 0.440854838
K11  (computationally) = 0.252212372
K11  (by hand) = 0.252212181
x2= 0.305470124
x1=  0.523374579
T by hand=147.726288
V by hand=-54.3831594
m2=11.912
Energy=  4196.76847
Energy without mass = 172.768465
x1=0.523374579
x2=0.305470124
x2bar=0.299230242
Kinetic Energy=147.726288
Potential Energy=-54.3831594
Volume Energy=79.4253359
```

$(f_2(x)/x)^{3/2}$

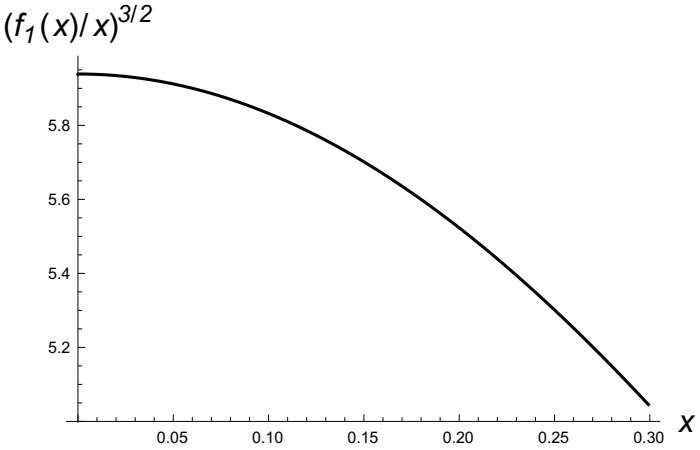
Out[ ]=



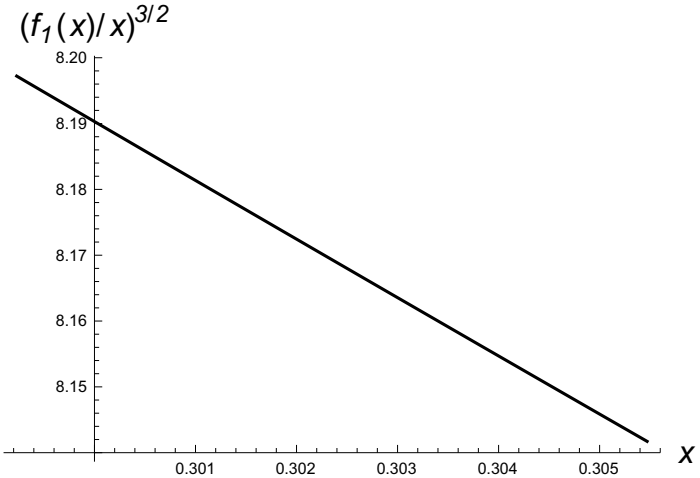
Out[ ]=


$(f_2(x)/x)^{3/2}$
32.95
32.90
32.85
32.80
0.301
0.302
0.303
0.304
0.305
x

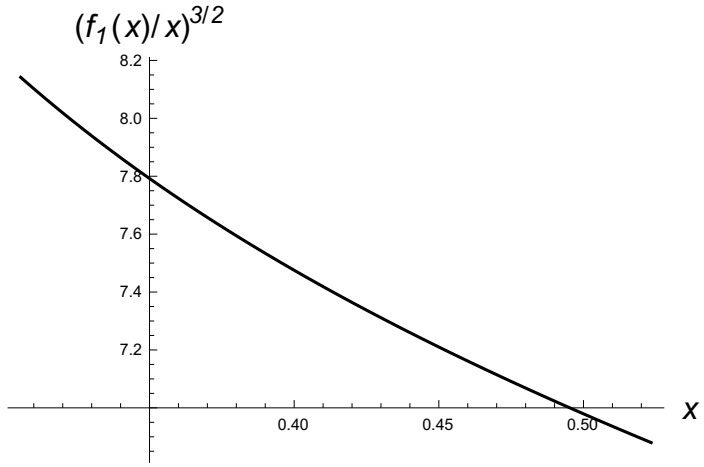
Out[ ]=


$(f_1(x)/x)^{3/2}$
5.8
5.6
5.4
5.2
0.05
0.10
0.15
0.20
0.25
0.30
x

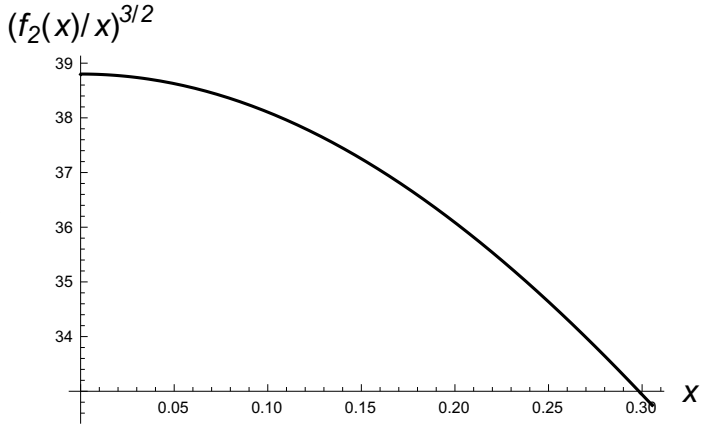
Out[ ]=


$(f_1(x)/x)^{3/2}$
8.20
8.19
8.18
8.17
8.16
8.15
0.301
0.302
0.303
0.304
0.305
x

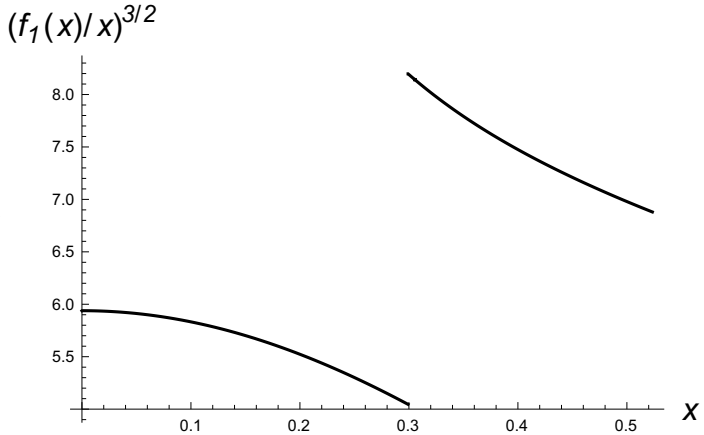
Out[ ]=


$(f_1(x)/x)^{3/2}$
8.2
8.0
7.8
7.6
7.4
7.2
0.40
0.45
0.50
x

Out[ ]=


$(f_2(x)/x)^{3/2}$
39
38
37
36
35
34
0.05
0.10
0.15
0.20
0.25
0.30
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
8.0
7.5
7.0
6.5
6.0
5.5
0.1
0.2
0.3
0.4
0.5
x

Out[ ]=

